

Strong Years program

# Back Strong



A back-friendly strength program to help you move with less stiffness: bend, carry, turn and walk with more confidence.

12 weeks · four levels · every session scripted.

Chang Yin & Sun Yoon

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Chang Yin and Sun Yoon are AI characters created by a team of people. Their life story is fiction. The exercises and recipes are built on published research for older adults (sources in the back). This is education, not medical advice. Check with your doctor before starting new exercise.

# Back Strong: how it works

## Chang Yin

I'm Chang. AI character. Twelve weeks, one program, done properly. A back-friendly strength program to help you move with less stiffness: bend, carry, turn and walk with more confidence.

A little more, slowly. That's the whole method.

**Who it's for.** Adults 60+ with an everyday stiff or achy back who are cleared for exercise. Not a treatment for any back condition.

**Ask your doctor first if:** Back pain after a fall, fever with back pain, night pain that wakes you and doesn't change with position, numbness in the saddle area, new leg weakness or loss of bladder or bowel control: those are for your doctor, today, not this program. Osteoporosis or a past spine fracture: you can do this program, following the long-back rule on every bend, and check with your doctor first.

**What you need.** Chair against a wall, kitchen counter, your bed, a grocery bag or jug, walking shoes. Optional: a resistance band with door anchor.

**Every day:** Stop if you feel chest pain or pressure, dizziness, unusual shortness of breath, or sharp pain. Chest pain, face drooping, arm weakness or trouble speaking? Call 911 now. Pain rule (E41): Mild discomfort up to 3 out of 10 is okay if it settles by tomorrow. Sharper, or worse the next day: back off and ask your doctor or physical therapist.

## Your week

Mon: program session A · Tue: mobility · Wed: program session B · Thu: strength · Fri: breath + qigong · Sat: walk-and-talk · Sun: rest + stretch. Program sessions are marked in **bold** in the plan; the other days use your Strong Years Daily Practice sessions (DP numbers).

**Walking goal each week:** wk 1: 10 min, 3 days · wk 2: 12 min, 3 days · wk 3: 15 min, 4 days · wk 4: 12 min, 3 days (easy week) · wk 5: 18 min, 4 days · wk 6: 20 min, 4 days · wk 7: 22 min, 5 days · wk 8: 18 min, 4 days (easy week) · wk 9: 25 min, 5 days · wk 10: 25 min, 5 days · wk 11: 30 min, 5 days · wk 12: 30 min, 5 days

## How it progresses

Three 4-week phases: **Foundation** (weeks 1–4, A1/B1), **Build** (weeks 5–8, A2/B2) and **Strengthen** (weeks 9–12, A3/B3).

| Week in each phase | Rule                                                                                                        |
|--------------------|-------------------------------------------------------------------------------------------------------------|
| 1st                | Doses as written in the session. Learn the moves. Stop each set with one or two good reps left in the tank. |

| Week in each phase | Rule                                                                                                               |
|--------------------|--------------------------------------------------------------------------------------------------------------------|
| 2nd                | Add 1–2 reps per set (never past 12). Holds up 5 seconds (never past 30).                                          |
| 3rd                | Where every rep felt easy two sessions in a row, use the Harder version or a heavier jug. Otherwise repeat week 2. |
| 4th (easy week)    | Easy week: same moves, one set fewer. Then do your checkpoint at the weekend.                                      |

**Levels.** Every session has four versions: Rebuild (chair-based), Steady, Strong and Iron. Move up only when every rep felt easy two sessions in a row. Move down any day. **Advanced** doses are never a starting point.

## The 12-week plan

| Week | Phase      | Mon           | Tue  | Wed           | Thu  | Fri  | Sat  | Sun  |
|------|------------|---------------|------|---------------|------|------|------|------|
| 1    | Foundation | <b>BKS-A1</b> | DP02 | <b>BKS-B1</b> | DP04 | DP05 | DP06 | DP07 |
| 2    | Foundation | <b>BKS-A1</b> | DP09 | <b>BKS-B1</b> | DP08 | DP12 | DP13 | DP14 |
| 3    | Foundation | <b>BKS-A1</b> | DP16 | <b>BKS-B1</b> | DP11 | DP19 | DP20 | DP21 |
| 4 ★  | Foundation | <b>BKS-A1</b> | DP23 | <b>BKS-B1</b> | DP15 | DP26 | DP27 | DP28 |
| 5    | Build      | <b>BKS-A2</b> | DP30 | <b>BKS-B2</b> | DP18 | DP05 | DP06 | DP07 |
| 6    | Build      | <b>BKS-A2</b> | DP02 | <b>BKS-B2</b> | DP22 | DP12 | DP13 | DP14 |
| 7    | Build      | <b>BKS-A2</b> | DP09 | <b>BKS-B2</b> | DP25 | DP19 | DP20 | DP21 |
| 8 ★  | Build      | <b>BKS-A2</b> | DP16 | <b>BKS-B2</b> | DP29 | DP26 | DP27 | DP28 |
| 9    | Strengthen | <b>BKS-A3</b> | DP23 | <b>BKS-B3</b> | DP01 | DP05 | DP06 | DP07 |
| 10   | Strengthen | <b>BKS-A3</b> | DP30 | <b>BKS-B3</b> | DP04 | DP12 | DP13 | DP14 |

| Week | Phase      | Mon           | Tue  | Wed           | Thu  | Fri  | Sat  | Sun  |
|------|------------|---------------|------|---------------|------|------|------|------|
| 11   | Strengthen | <b>BKS-A3</b> | DP02 | <b>BKS-B3</b> | DP08 | DP19 | DP20 | DP21 |
| 12 ★ | Strengthen | <b>BKS-A3</b> | DP09 | <b>BKS-B3</b> | DP11 | DP26 | DP27 | DP28 |

★ = easy week, then checkpoint. **Bold** = program session. DP = Daily Practice session number (sessions.json).

**Red flags:** Not for exercise. For your doctor, today: chest pain or pressure; fainting; sudden weakness or numbness in the face, arm or leg; trouble speaking; a sudden severe headache; new confusion; one calf that is swollen, red and painful; a hot, swollen joint; night pain that wakes you and doesn't change with position; fever with back pain; loss of bladder or bowel control or numbness in the saddle area; a fall where you hit your head (especially on blood thinners). Chest pain, face drooping, arm weakness or trouble speaking? Call 911 now.

# Your checkpoints

Start, then the end of weeks 4, 8 and 12. Same chair, same shoes, same time of day. These are fitness checks from published research, **not medical tests**.

| Checkpoint                             | How                                                                            | Start | Week 4 | Week 8 | Week 12 |
|----------------------------------------|--------------------------------------------------------------------------------|-------|--------|--------|---------|
| <b>Morning stiffness</b> (self-rating) | 0-10, first 10 minutes after getting up (0 = none)                             |       |        |        |         |
| <b>Sock test</b> (self-rating)         | Putting on socks seated: easy / some effort / hard / need help                 |       |        |        |         |
| <b>Jug pick-up comfort</b> (E40, E41)  | Hinge a half-gallon jug from a chair seat and back, 5 times: discomfort 0-10   |       |        |        |         |
| <b>Comfortable walking</b> (E60, E22)  | Minutes you can walk at an easy pace before stiffness makes you stop, up to 30 |       |        |        |         |
| <b>30-second chair stand</b> (E11)     | Full stands                                                                    |       |        |        |         |

# The program sessions

Full follow-along scripts for the six program sessions. The same content is in programs.json for the app and video pipeline.

# BKS-A1 • Phase 1 (Foundation): Hinge and brace

| Type                | Length | Set         | Wardrobe  | Voices |
|---------------------|--------|-------------|-----------|--------|
| Back Strong program | 10:05  | SET-KITCHEN | C-TRAIN-B | Chang  |

**On screen the whole session:** the "AI character" corner tag. In the first 30 seconds and before each hard block: "Stop if chest pain, dizziness or sharp pain."

**Gentle-day swaps (the sore knee / sore back / low energy button):** **Sore knee:** Use the Rebuild version of any standing leg move; skip floor practice and stairs today. **Sore back:** Seated versions of hinges and carries; keep every bend long-backed. **Low energy:** Do the warm-up, the first two blocks and the cool-down.

## Run of show

| Time      | Segment                                       | Rebuild                            | Steady     | Strong     | Iron       |
|-----------|-----------------------------------------------|------------------------------------|------------|------------|------------|
| 0:00–0:35 | <b>Welcome</b>                                | All levels together                |            |            |            |
| 0:35–1:00 | <b>Ankle pumps</b><br>Ankle pumps and circles | 10 pumps                           | 10 pumps   | 10 pumps   | 10 pumps   |
| 1:00–1:40 | <b>March</b><br>Standing march at the counter | <i>Seated march:</i><br>40 seconds | 40 seconds | 40 seconds | 40 seconds |

| Time      | Segment                                                        | Rebuild                                    | Steady                            | Strong                      | Iron                            |
|-----------|----------------------------------------------------------------|--------------------------------------------|-----------------------------------|-----------------------------|---------------------------------|
| 1:40–2:00 | <b>Shoulder rolls</b>                                          | 6 rolls                                    | 8 rolls                           | 8 rolls                     | 8 rolls + circles               |
| 2:00–3:00 | <b>Seated hinge</b><br>Seated hip hinge                        | 6 small hinges                             | 8 hinges                          | 10 hinges                   | 10 hinges, bottle at chest      |
| 3:00–4:20 | <b>Counter hip hinge</b>                                       | <i>Seated hip hinge</i> : 2 sets of 8      | 2 sets of 8, hands on the counter | 2 sets of 10                | 3 sets of 8, jug from the chair |
| 4:20–5:30 | <b>Standing bird-dog</b><br>Standing bird-dog at the counter   | 4 each side, leg only                      | 6 each side                       | 8 each side, 3-second hold  | 10 each side, 3-second hold     |
| 5:30–6:30 | <b>Side lean at the counter</b>                                | <i>Seated chest lift</i> : 8 lifts, seated | 2 × 5 seconds each side           | 2 × 10 seconds each side    | 3 × 15 seconds each side        |
| 6:30–7:30 | <b>Bridge on the bed</b>                                       | 6 squeezes                                 | 2 sets of 8                       | 2 sets of 10, 3-second hold | 3 sets of 10, 3-second hold     |
| 7:30–8:30 | <b>Clamshell on the bed</b><br>Side-lying clamshell on the bed | 6 each side, small                         | 8 each side                       | 10 each side                | 10 each side, band              |
| 8:30–9:00 | <b>Stand up slowly</b><br>Ankle pumps and circles              | 10 pumps                                   | 10 pumps                          | 10 pumps                    | 10 pumps                        |
| 9:00–9:40 | <b>4-6 breathing</b>                                           | 40 seconds                                 | 40 seconds                        | 40 seconds                  | 40 seconds                      |

| Time       | Segment      | Rebuild             | Steady | Strong | Iron |
|------------|--------------|---------------------|--------|--------|------|
| 9:40–10:05 | <b>Close</b> | All levels together |        |        |      |

## Script

### 0:00–0:35 • Welcome

| Time | Who   | Spoken                                                                                         | On screen                                   |
|------|-------|------------------------------------------------------------------------------------------------|---------------------------------------------|
| 0:00 | Chang | Back Strong. Phase one: hinge and brace.                                                       | Back Strong · A1                            |
| 0:10 | Chang | Our goal is moving with less stiffness. A stronger, more confident back.                       | Stop if chest pain, dizziness or sharp pain |
| 0:17 | Chang | Pain rule: up to three out of ten, settled by tomorrow, is okay. Worse the next day, back off. |                                             |
| 0:24 | Chang | Chest pain, dizziness or sharp pain: stop, sit, and call your doctor.                          |                                             |

### 0:35–1:00 • Ankle pumps

| Time | Who   | Spoken                                                            | On screen                 |
|------|-------|-------------------------------------------------------------------|---------------------------|
| 0:35 | Chang | Sit tall. Chair against the wall.                                 | Ankle pumps               |
| 0:42 | Chang | Point your toes away, then pull them toward you. That's one pump. | Easier: Smaller movements |
| 0:46 | Chang | Big, slow movements.                                              |                           |
| 0:50 | Chang | Feel your calves work.                                            |                           |
| 0:54 | Chang | Breathe slowly.                                                   |                           |

**Cues:** Big, slow movements. Feel your calves work. **Easier:** Smaller movements. **Harder:** Lift one foot off the floor while you circle.

**Safety callouts:** Great before you stand up from sitting or lying. Sit a moment before you stand, then stand slowly while holding something steady.

## 1:00–1:40 • March

| Time | Who   | Spoken                              | On screen             |
|------|-------|-------------------------------------|-----------------------|
| 1:00 | Chang | Stand slowly. Hand on the counter.  | March                 |
| 1:19 | Chang | March. Easy rhythm. Breathe steady. | Easier: Sit and march |

**Cues:** Stand tall. Knee up, foot down softly. **Easier:** Sit and do the seated march. **Harder:** Lift knees higher, march faster, or let one hand hover above the counter.

## 1:40–2:00 • Shoulder rolls

| Time | Who   | Spoken                                        | On screen                     |
|------|-------|-----------------------------------------------|-------------------------------|
| 1:40 | Chang | Sit tall. Chair against the wall.             | Shoulder rolls                |
| 1:46 | Chang | Lift your shoulders up toward your ears.      | Easier: Just shrug and let go |
| 1:49 | Chang | Big, slow circles, backward.                  |                               |
| 1:52 | Chang | Let your arms hang heavy.                     |                               |
| 1:55 | Chang | Breathe in as you lift, out as you roll down. |                               |

**Cues:** Big, slow circles, backward. Let your arms hang heavy. **Easier:** Just shrug up and let go. **Harder:** Add arm circles: small circles out to the sides, forward and back.

## 2:00–3:00 • Seated hinge

| Time | Who   | Spoken                                                                                              | On screen                   |
|------|-------|-----------------------------------------------------------------------------------------------------|-----------------------------|
| 2:00 | Chang | Sit tall. Chair against the wall.                                                                   | Seated hinge                |
| 2:12 | Chang | Keep your back long and tip your chest forward from the hips, sliding your hands toward your knees. | Easier: Smaller tip forward |
| 2:22 | Chang | Long back, proud chest.                                                                             |                             |
| 2:31 | Chang | Bend at the hips, not the belly.                                                                    |                             |
| 2:41 | Chang | Breathe out as you sit tall.                                                                        |                             |

**Cues:** Long back, proud chest. Bend at the hips, not the belly. **Easier:** A smaller tip forward. **Harder:** Hold a water bottle against your chest.

**Safety callouts:** Osteoporosis: this is the safe bending pattern. Keep the back long, never curl.

## 3:00–4:20 • Counter hip hinge

| Time | Who   | Spoken                                                         | On screen                              |
|------|-------|----------------------------------------------------------------|----------------------------------------|
| 3:00 | Chang | Hand on the counter. Chair behind you.                         | Counter hip hinge                      |
| 3:15 | Chang | Push your hips back, like closing a car door with your bottom. | Easier: Smaller hinge, hands stay down |
| 3:28 | Chang | Hips go back, not down.                                        |                                        |
| 3:36 | Chang | One... two... breathe out... three... slow down... four.       |                                        |
| 3:41 | Chang | Long back, chest proud. No rounding.                           |                                        |
| 3:54 | Chang | Breathe out as you stand tall.                                 |                                        |

**Cues:** Hips go back, not down. Long back, chest proud. No rounding. Stand up by squeezing your buttocks. **Easier:** Smaller movement, hands stay on the counter the whole time. **Harder:** Pick up a jug of water from the seat of a chair using the same hinge, and set it back down. Later: from a lower surface, like a step.

**Safety callouts:** Osteoporosis: this is the safe bending pattern. Keep the back long, never curl. Osteoporosis or a past spine fracture: this is the safe way to bend. Keep your back long; never round forward under load.

## 4:20–5:30 • Standing bird-dog

| Time | Who   | Spoken                                                                                         | On screen                         |
|------|-------|------------------------------------------------------------------------------------------------|-----------------------------------|
| 4:20 | Chang | Hand on the counter. Chair behind you.                                                         | Standing bird-dog                 |
| 4:34 | Chang | Slide your right foot back and lift it a few inches behind you, toes on or just off the floor. | Easier: Leg only, both hands down |
| 4:45 | Chang | Long back. No arching.                                                                         |                                   |
| 4:56 | Chang | Squeeze the buttock of the moving leg.                                                         |                                   |
| 5:07 | Chang | Breathe out as you reach.                                                                      |                                   |

**Cues:** Long back. No arching. Squeeze the buttock of the moving leg.

**Easier:** Leg only; both hands stay on the counter. **Harder:** Hold the reach for three seconds.

## 5:30–6:30 • Side lean at the counter

| Time | Who   | Spoken                                 | On screen                |
|------|-------|----------------------------------------|--------------------------|
| 5:30 | Chang | Hand on the counter. Chair behind you. | Side lean at the counter |

| Time | Who   | Spoken                                                              | On screen                      |
|------|-------|---------------------------------------------------------------------|--------------------------------|
| 5:42 | Chang | Lean your body into your forearm in a straight line, feet together. | Easier: Closer to counter, 5 s |
| 5:52 | Chang | Straight line from head to heels.                                   |                                |
| 6:01 | Chang | Keep talking. Your breathing never stops.                           |                                |
| 6:11 | Chang | Breathe normally the whole time. Count out loud.                    |                                |

**Cues:** Straight line from head to heels. Keep talking. Your breathing never stops. **Easier:** Stand closer to the counter, or hold for 5 seconds. **Harder:** Step farther away, or hold up to 20 seconds.

**Safety callouts:** Breathe through it. High blood pressure or a heart condition? Ask your doctor before long holds.

## 6:30–7:30 • Bridge on the bed

| Time | Who   | Spoken                                                   | On screen                       |
|------|-------|----------------------------------------------------------|---------------------------------|
| 6:30 | Chang | On your bed. Sit down first, then lie back slowly.       | Bridge on the bed               |
| 6:42 | Chang | Squeeze your buttocks and lift your hips a few inches.   | Easier: Squeeze without lifting |
| 6:52 | Chang | Push through your heels.                                 |                                 |
| 6:57 | Chang | One... two... breathe out... three... slow down... four. |                                 |
| 7:01 | Chang | Lift with your buttocks, not your back.                  |                                 |
| 7:11 | Chang | Breathe out as you lift.                                 |                                 |

**Cues:** Push through your heels. Lift with your buttocks, not your back.

**Easier:** Just squeeze your buttocks without lifting. **Harder:** Hold three seconds at the top, or lift higher.

**Safety callouts:** Sit a moment before you stand, then stand slowly while holding something steady.

## 7:30–8:30 • Clamshell on the bed

| Time | Who   | Spoken                                                       | On screen            |
|------|-------|--------------------------------------------------------------|----------------------|
| 7:30 | Chang | On your bed. Sit down first, then lie back slowly.           | Clamshell on the bed |
| 7:42 | Chang | Keeping your feet together, lift your top knee a few inches. | Easier: Smaller lift |
| 7:52 | Chang | Hips stay stacked. Don't roll back.                          |                      |
| 8:01 | Chang | Small and controlled.                                        |                      |
| 8:11 | Chang | Breathe out as you lift.                                     |                      |

**Cues:** Hips stay stacked. Don't roll back. Small and controlled. **Easier:** Smaller lift. **Harder:** Loop a light band above your knees.

**Safety callouts:** New hip? Side-lying may not be allowed yet; ask your surgeon. Sit a moment before you stand, then stand slowly while holding something steady.

## 8:30–9:00 • Stand up slowly

| Time | Who   | Spoken                                                               | On screen                 |
|------|-------|----------------------------------------------------------------------|---------------------------|
| 8:30 | Chang | Roll to your side. Push up to sitting. Ankle pumps.                  | Stand up slowly           |
| 8:41 | Chang | Sit a moment before you stand. Then stand slowly, holding something. | Easier: Smaller movements |

| Time | Who   | Spoken          | On screen                     |
|------|-------|-----------------|-------------------------------|
| 8:49 | Chang | Breathe slowly. | Sit a moment before you stand |

**Cues:** Big, slow movements. Feel your calves work. **Easier:** Smaller movements. **Harder:** Lift one foot off the floor while you circle.

**Safety callouts:** Great before you stand up from sitting or lying. Sit a moment before you stand, then stand slowly while holding something steady.

## 9:00–9:40 • 4-6 breathing

| Time | Who   | Spoken                                                           | On screen                   |
|------|-------|------------------------------------------------------------------|-----------------------------|
| 9:00 | Chang | Sit tall. Chair against the wall.                                | 4-6 breathing               |
| 9:09 | Chang | Breathe in through your nose for a count of 4.                   | Easier: In for 3, out for 4 |
| 9:15 | Chang | Belly rises first.                                               |                             |
| 9:22 | Chang | Soft shoulders.                                                  |                             |
| 9:28 | Chang | In for 4, out for 6. That is about 6 breaths a minute. No holds. |                             |

**Cues:** Belly rises first. Soft shoulders. **Easier:** In for 3, out for 4. **Harder:** Continue for 5 minutes.

**Safety callouts:** If you feel light-headed, go back to normal breathing.

## 9:40–10:05 • Close

| Time | Who   | Spoken                                         | On screen |
|------|-------|------------------------------------------------|-----------|
| 9:40 | Chang | Long back, strong hips. That's the foundation. |           |

| Time | Who   | Spoken                                              | On screen |
|------|-------|-----------------------------------------------------|-----------|
| 9:53 | Chang | Walk today: check your walking minutes on the plan. |           |

# BKS-B1 • Phase 1 (Foundation):

## Turn and carry

| Type                | Length | Set        | Wardrobe | Voices |
|---------------------|--------|------------|----------|--------|
| Back Strong program | 9:30   | SET-LIVING | C-CASUAL | Chang  |

**On screen the whole session:** the "AI character" corner tag. In the first 30 seconds and before each hard block: "Stop if chest pain, dizziness or sharp pain."

**Gentle-day swaps (the sore knee / sore back / low energy button):** **Sore knee:** Use the Rebuild version of any standing leg move; skip floor practice and stairs today. **Sore back:** Seated versions of hinges and carries; keep every bend long-backed. **Low energy:** Do the warm-up, the first two blocks and the cool-down.

## Run of show

| Time      | Segment                       | Rebuild             | Steady      | Strong                  | Iron                    |
|-----------|-------------------------------|---------------------|-------------|-------------------------|-------------------------|
| 0:00–0:35 | <b>Welcome</b>                | All levels together |             |                         |                         |
| 0:35–1:25 | <b>Pelvic tilt on the bed</b> | 6 small tilts       | 8 tilts     | 10 tilts, 2-second hold | 10 tilts, 5-second hold |
| 1:25–2:15 | <b>Knee rocks on the bed</b>  | 4 each side, tiny   | 6 each side | 8 each side             | 8 each side, pause      |

| Time          | Segment                                              | Rebuild                                      | Steady                   | Strong                   | Iron                                 |
|---------------|------------------------------------------------------|----------------------------------------------|--------------------------|--------------------------|--------------------------------------|
| 2:15–<br>2:45 | <b>Stand up slowly</b><br>Ankle pumps and circles    | 10 pumps                                     | 10 pumps                 | 10 pumps                 | 10 pumps                             |
| 2:45–<br>3:30 | <b>Chest lift</b><br>Seated chest lift               | 6 lifts                                      | 8 lifts                  | 8 lifts, 3-second hold   | 10 lifts, 3-second hold              |
| 3:30–<br>4:15 | <b>Wall slides</b>                                   | 6 finger walks                               | 8 slides                 | 10 slides                | 10 slides, 2-second hold             |
| 4:15–<br>5:25 | <b>Band press-out</b><br>Band press-out (anti-twist) | 2 sets of 6 each side, seated                | 2 sets of 8 each side    | 3 sets of 8 each side    | 3 sets of 8 each side, 5-second hold |
| 5:25–<br>6:40 | <b>One-side carry</b>                                | <i>Towel wring (grip):</i> 3 wrings each way | 2 × 20 seconds each side | 2 × 30 seconds each side | 3 × 30 seconds each side, heavier    |
| 6:40–<br>7:40 | <b>Standing hip-front stretch</b>                    | 20 seconds each side, both hands down        | 2 × 20 seconds each side | 2 × 20 seconds each side | 2 × 30 seconds each side, arm up     |
| 7:40–<br>8:25 | <b>Gentle seated turn</b>                            | 4 each side, small                           | 6 each side              | 6 each side              | 8 each side                          |
| 8:25–<br>9:05 | <b>Cyclic sighing</b>                                | 40 seconds                                   | 40 seconds               | 40 seconds               | 40 seconds                           |
| 9:05–<br>9:30 | <b>Close</b>                                         | All levels together                          |                          |                          |                                      |

# Script

## 0:00–0:35 • Welcome

| Time | Who   | Spoken                                                                | On screen                                   |
|------|-------|-----------------------------------------------------------------------|---------------------------------------------|
| 0:00 | Chang | Back Strong. Turn and carry.                                          | Back Strong · B1                            |
| 0:10 | Chang | Gentle turning, then carrying on one side, standing tall.             | Stop if chest pain, dizziness or sharp pain |
| 0:17 | Chang | Osteoporosis: turns stay small and gentle. Never force a twist.       |                                             |
| 0:24 | Chang | Chest pain, dizziness or sharp pain: stop, sit, and call your doctor. |                                             |

## 0:35–1:25 • Pelvic tilt on the bed

| Time | Who   | Spoken                                                                     | On screen                        |
|------|-------|----------------------------------------------------------------------------|----------------------------------|
| 0:35 | Chang | On your bed. Sit down first, then lie back slowly.                         | Pelvic tilt on the bed           |
| 0:46 | Chang | Gently flatten your lower back into the mattress by tightening your belly. | Easier: Smaller, one-second hold |
| 0:54 | Chang | Small movement.                                                            |                                  |
| 1:02 | Chang | Belly tightens, bottom stays down.                                         |                                  |
| 1:10 | Chang | Breathe out as you flatten.                                                |                                  |

**Cues:** Small movement. Belly tightens, bottom stays down. **Easier:** An even smaller movement, one-second hold. **Harder:** Hold five seconds.

**Safety callouts:** Osteoporosis or a past spine fracture: keep it small and gentle. Sit a moment before you stand, then stand slowly while holding something steady.

## 1:25–2:15 • Knee rocks on the bed

| Time | Who   | Spoken                                                                | On screen               |
|------|-------|-----------------------------------------------------------------------|-------------------------|
| 1:25 | Chang | On your bed. Sit down first, then lie back slowly.                    | Knee rocks on the bed   |
| 1:36 | Chang | Let both knees drift gently to the right, only as far as comfortable. | Easier: A tiny movement |
| 1:44 | Chang | Small and slow.                                                       |                         |
| 1:52 | Chang | Shoulders stay on the bed.                                            |                         |
| 2:00 | Chang | Breathe out as the knees drift.                                       |                         |

**Cues:** Small and slow. Shoulders stay on the bed. **Easier:** A tiny movement, a few inches. **Harder:** Pause for one slow breath on each side.

**Safety callouts:** Osteoporosis? Turn gently and only as far as comfortable. Never force the twist. Sit a moment before you stand, then stand slowly while holding something steady.

## 2:15–2:45 • Stand up slowly

| Time | Who   | Spoken                                              | On screen                     |
|------|-------|-----------------------------------------------------|-------------------------------|
| 2:15 | Chang | Roll to your side, push up to sitting. Ankle pumps. | Stand up slowly               |
| 2:26 | Chang | Sit a moment, then stand slowly, holding something. | Easier: Smaller movements     |
| 2:34 | Chang | Breathe slowly.                                     | Sit a moment before you stand |

**Cues:** Big, slow movements. Feel your calves work. **Easier:** Smaller movements. **Harder:** Lift one foot off the floor while you circle.

**Safety callouts:** Great before you stand up from sitting or lying. Sit a moment before you stand, then stand slowly while holding something steady.

## 2:45–3:30 • Chest lift

| Time | Who   | Spoken                                               | On screen            |
|------|-------|------------------------------------------------------|----------------------|
| 2:45 | Chang | Sit tall. Chair against the wall.                    | Chest lift           |
| 2:55 | Chang | Lift your chest up toward the ceiling, growing tall. | Easier: Smaller lift |
| 3:02 | Chang | Grow tall.                                           |                      |
| 3:09 | Chang | Look straight ahead, not up.                         |                      |
| 3:16 | Chang | Breathe in as you lift, out as you relax.            |                      |

**Cues:** Grow tall. Look straight ahead, not up. **Easier:** Smaller lift. **Harder:** Hold three seconds at the top.

## 3:30–4:15 • Wall slides

| Time | Who   | Spoken                                                  | On screen                           |
|------|-------|---------------------------------------------------------|-------------------------------------|
| 3:30 | Chang | Wall in front of you or behind you. Chair nearby.       | Wall slides                         |
| 3:40 | Chang | Slide your arms up the wall only as far as comfortable. | Easier: Walk fingertips up the wall |
| 3:47 | Chang | Ribs down.                                              |                                     |
| 3:54 | Chang | Slow and smooth.                                        |                                     |

| Time | Who   | Spoken                                             | On screen |
|------|-------|----------------------------------------------------|-----------|
| 4:01 | Chang | Breathe in as you slide up, out as you slide down. |           |

**Cues:** Ribs down. Slow and smooth. **Easier:** Face the wall and walk your fingertips up it instead. **Harder:** Hold for two seconds at the top.

**Safety callouts:** Shoulder pain or a known rotator cuff problem? Only lift to a comfortable height, or skip it.

### 4:15–5:25 • Band press-out

| Time | Who   | Spoken                                                        | On screen                          |
|------|-------|---------------------------------------------------------------|------------------------------------|
| 4:15 | Chang | Sit or stand with the chair behind you. Check the band first. | Band press-out                     |
| 4:29 | Chang | Press your hands straight out in front of your chest.         | Easier: Seated, closer to the door |
| 4:40 | Chang | Stay square. The band pulls, you don't turn.                  |                                    |
| 4:46 | Chang | One... two... breathe out... three... slow down... four.      |                                    |
| 4:51 | Chang | Ribs down.                                                    |                                    |
| 5:02 | Chang | Breathe out as you press.                                     |                                    |

**Cues:** Stay square. The band pulls, you don't turn. Ribs down. **Easier:** Do it seated, or stand closer to the door. **Harder:** Step farther from the door, or hold 5 seconds.

**Safety callouts:** Door must open away from you and be locked.

## 5:25–6:40 • One-side carry

| Time | Who   | Spoken                                          | On screen                            |
|------|-------|-------------------------------------------------|--------------------------------------|
| 5:25 | Chang | Clear path. Free hand trails along the counter. | One-side carry                       |
| 5:40 | Chang | Stand tall. Don't lean away from the bag.       | Easier: Lighter bag, hand on counter |
| 5:52 | Chang | Tall like a lamppost.                           |                                      |
| 6:04 | Chang | Shoulders level.                                |                                      |
| 6:16 | Chang | Breathe steadily. You should be able to talk.   |                                      |

**Cues:** Tall like a lamppost. Shoulders level. **Easier:** A lighter bag, free hand on the counter the whole time. **Harder:** A heavier bag, or a longer walk.

**Safety callouts:** Hand arthritis flaring today? Squeeze gently, or skip and do the finger spread only. Pick up and put down with a hinge. Never round your back to lift.

## 6:40–7:40 • Standing hip-front stretch

| Time | Who   | Spoken                                                                     | On screen                             |
|------|-------|----------------------------------------------------------------------------|---------------------------------------|
| 6:40 | Chang | Hand on the counter. Chair behind you.                                     | Standing hip-front stretch            |
| 6:52 | Chang | Tuck your tailbone under slightly and squeeze the buttock of the back leg. | Easier: Smaller step, both hands down |
| 7:02 | Chang | Stay tall. Don't arch your back.                                           |                                       |
| 7:11 | Chang | Small movement, big squeeze.                                               |                                       |
| 7:21 | Chang | Breathe slowly.                                                            |                                       |

**Cues:** Stay tall. Don't arch your back. Small movement, big squeeze.

**Easier:** Smaller step. **Harder:** Reach the arm on the back-leg side up toward the ceiling, to comfortable height.

## 7:40–8:25 • Gentle seated turn

| Time | Who   | Spoken                                                                 | On screen             |
|------|-------|------------------------------------------------------------------------|-----------------------|
| 7:40 | Chang | Sit tall. Chair against the wall.                                      | Gentle seated turn    |
| 7:50 | Chang | Turn your chest to the right, only as far as comfortable. Eyes follow. | Easier: Smaller turns |
| 7:57 | Chang | Grow tall first, then turn.                                            |                       |
| 8:04 | Chang | Hips stay still.                                                       |                       |
| 8:11 | Chang | Breathe out as you turn.                                               |                       |

**Cues:** Grow tall first, then turn. Hips stay still. **Easier:** Smaller turns.

**Harder:** Place one hand on the opposite knee as a light guide. Never pull into the turn.

**Safety callouts:** Osteoporosis? Turn gently and only as far as comfortable. Never force the twist.

## 8:25–9:05 • Cyclic sighing

| Time | Who   | Spoken                            | On screen                        |
|------|-------|-----------------------------------|----------------------------------|
| 8:25 | Chang | Sit tall. Chair against the wall. | Cyclic sighing                   |
| 8:34 | Chang | Breathe in through your nose.     | Easier: Just slow nose breathing |
| 8:40 | Chang | Two sniffs in, one long sigh out. |                                  |
| 8:47 | Chang | No holding at any point.          |                                  |

| Time | Who   | Spoken                                                               | On screen |
|------|-------|----------------------------------------------------------------------|-----------|
| 8:53 | Chang | The out-breath is always longer than the in-breath. No breath holds. |           |

**Cues:** Two sniffs in, one long sigh out. No holding at any point. **Easier:** Just breathe slowly through your nose, making the out-breath a little longer. **Harder:** Build up to 5 minutes.

**Safety callouts:** If you feel light-headed, return to normal breathing.

## 9:05–9:30 • Close

| Time | Who   | Spoken                                               | On screen |
|------|-------|------------------------------------------------------|-----------|
| 9:05 | Chang | Stand tall. Turn gently. Carry proud.                |           |
| 9:18 | Chang | Walk today, and log your stiffness tomorrow morning. |           |

# BKS-A2 • Phase 2 (Build): Hinge with load

| Type                | Length | Set         | Wardrobe  | Voices |
|---------------------|--------|-------------|-----------|--------|
| Back Strong program | 9:15   | SET-KITCHEN | C-TRAIN-B | Chang  |

**On screen the whole session:** the "AI character" corner tag. In the first 30 seconds and before each hard block: "Stop if chest pain, dizziness or sharp pain."

**Gentle-day swaps (the sore knee / sore back / low energy button):** **Sore knee:** Use the Rebuild version of any standing leg move; skip floor practice and stairs today. **Sore back:** Seated versions of hinges and carries; keep every bend long-backed. **Low energy:** Do the warm-up, the first two blocks and the cool-down.

## Run of show

| Time      | Segment                                       | Rebuild             | Steady   | Strong   | Iron     |
|-----------|-----------------------------------------------|---------------------|----------|----------|----------|
| 0:00–0:35 | <b>Welcome</b>                                | All levels together |          |          |          |
| 0:35–1:00 | <b>Ankle pumps</b><br>Ankle pumps and circles | 10 pumps            | 10 pumps | 10 pumps | 10 pumps |

| Time          | Segment                                                      | Rebuild                                               | Steady                                       | Strong                          | Iron                                 |
|---------------|--------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------|---------------------------------|--------------------------------------|
| 1:00–<br>1:40 | <b>March</b><br>Standing march at the counter                | <i>Seated march:</i><br>40 seconds                    | 40 seconds                                   | 40 seconds                      | 40 seconds                           |
| 1:40–<br>2:00 | <b>Shoulder rolls</b>                                        | 6 rolls                                               | 8 rolls                                      | 8 rolls                         | 8 rolls + circles                    |
| 2:00–<br>3:10 | <b>Counter hip hinge</b>                                     | <i>Seated hip hinge:</i> 2 sets of 8, bottle at chest | 3 sets of 8                                  | 3 sets of 8, jug from the chair | 3 sets of 10, jug from the step      |
| 3:10–<br>4:30 | <b>Suitcase lift</b><br>Suitcase lift (one-hand hinge)       | <i>Seated hip hinge:</i> 2 sets of 8                  | 2 sets of 6 each side, from a higher surface | 2 sets of 8 each side           | 3 sets of 8 each side, from the step |
| 4:30–<br>5:30 | <b>Standing bird-dog</b><br>Standing bird-dog at the counter | 6 each side, leg only                                 | 8 each side, 3-second hold                   | 10 each side, 3-second hold     | 12 each side, 3-second hold          |
| 5:30–<br>6:30 | <b>Side lean at the counter</b>                              | <i>Seated chest lift:</i> 8 lifts, 3-second hold      | 2 × 10 seconds each side                     | 3 × 15 seconds each side        | 3 × 20 seconds each side             |
| 6:30–<br>7:30 | <b>Bridge on the bed</b>                                     | 8 squeezes                                            | 2 sets of 10                                 | 3 sets of 10, 3-second hold     | 3 sets of 12, 3-second hold          |
| 7:30–<br>8:00 | <b>Stand up slowly</b><br>Ankle pumps and circles            | 10 pumps                                              | 10 pumps                                     | 10 pumps                        | 10 pumps                             |
| 8:00–<br>8:50 | <b>4-6 breathing</b>                                         | 50 seconds                                            | 50 seconds                                   | 50 seconds                      | 50 seconds                           |

| Time      | Segment | Rebuild             | Steady | Strong | Iron |
|-----------|---------|---------------------|--------|--------|------|
| 8:50–9:15 | Close   | All levels together |        |        |      |

## Script

### 0:00–0:35 • Welcome

| Time | Who   | Spoken                                                                      | On screen                                   |
|------|-------|-----------------------------------------------------------------------------|---------------------------------------------|
| 0:00 | Chang | Phase two. We add load to the hinge.                                        | Back Strong · A2                            |
| 0:10 | Chang | The long back stays. The jug gets heavier.                                  | Stop if chest pain, dizziness or sharp pain |
| 0:17 | Chang | Pain above three out of ten that doesn't settle by morning: go back a step. |                                             |
| 0:24 | Chang | Chest pain, dizziness or sharp pain: stop, sit, and call your doctor.       |                                             |

### 0:35–1:00 • Ankle pumps

| Time | Who   | Spoken                                                            | On screen                 |
|------|-------|-------------------------------------------------------------------|---------------------------|
| 0:35 | Chang | Sit tall. Chair against the wall.                                 | Ankle pumps               |
| 0:42 | Chang | Point your toes away, then pull them toward you. That's one pump. | Easier: Smaller movements |
| 0:46 | Chang | Big, slow movements.                                              |                           |
| 0:50 | Chang | Feel your calves work.                                            |                           |
| 0:54 | Chang | Breathe slowly.                                                   |                           |

**Cues:** Big, slow movements. Feel your calves work. **Easier:** Smaller movements. **Harder:** Lift one foot off the floor while you circle.

**Safety callouts:** Great before you stand up from sitting or lying. Sit a moment before you stand, then stand slowly while holding something steady.

## 1:00–1:40 • March

| Time | Who   | Spoken                              | On screen             |
|------|-------|-------------------------------------|-----------------------|
| 1:00 | Chang | Stand slowly. Hand on the counter.  | March                 |
| 1:19 | Chang | March. Easy rhythm. Breathe steady. | Easier: Sit and march |

**Cues:** Stand tall. Knee up, foot down softly. **Easier:** Sit and do the seated march. **Harder:** Lift knees higher, march faster, or let one hand hover above the counter.

## 1:40–2:00 • Shoulder rolls

| Time | Who   | Spoken                                        | On screen                     |
|------|-------|-----------------------------------------------|-------------------------------|
| 1:40 | Chang | Sit tall. Chair against the wall.             | Shoulder rolls                |
| 1:46 | Chang | Lift your shoulders up toward your ears.      | Easier: Just shrug and let go |
| 1:49 | Chang | Big, slow circles, backward.                  |                               |
| 1:52 | Chang | Let your arms hang heavy.                     |                               |
| 1:55 | Chang | Breathe in as you lift, out as you roll down. |                               |

**Cues:** Big, slow circles, backward. Let your arms hang heavy. **Easier:** Just shrug up and let go. **Harder:** Add arm circles: small circles out to the sides, forward and back.

## 2:00–3:10 • Counter hip hinge

| Time | Who   | Spoken                                                         | On screen                              |
|------|-------|----------------------------------------------------------------|----------------------------------------|
| 2:00 | Chang | Hand on the counter. Chair behind you.                         | Counter hip hinge                      |
| 2:14 | Chang | Push your hips back, like closing a car door with your bottom. | Easier: Smaller hinge, hands stay down |
| 2:25 | Chang | Hips go back, not down.                                        |                                        |
| 2:31 | Chang | One... two... breathe out... three... slow down... four.       |                                        |
| 2:36 | Chang | Long back, chest proud. No rounding.                           |                                        |
| 2:47 | Chang | Breathe out as you stand tall.                                 |                                        |

**Cues:** Hips go back, not down. Long back, chest proud. No rounding. Stand up by squeezing your buttocks. **Easier:** Smaller movement, hands stay on the counter the whole time. **Harder:** Pick up a jug of water from the seat of a chair using the same hinge, and set it back down. Later: from a lower surface, like a step.

**Safety callouts:** Osteoporosis: this is the safe bending pattern. Keep the back long, never curl. Osteoporosis or a past spine fracture: this is the safe way to bend. Keep your back long; never round forward under load.

## 3:10–4:30 • Suitcase lift

| Time | Who   | Spoken                                                             | On screen                           |
|------|-------|--------------------------------------------------------------------|-------------------------------------|
| 3:10 | Chang | Hand on the counter. Chair behind you.                             | Suitcase lift                       |
| 3:25 | Chang | Push your hips back with a long back and bend your knees a little. | Easier: Lighter jug, higher surface |
| 3:38 | Chang | Hips back. Long back.                                              |                                     |

| Time | Who   | Spoken                                                   | On screen |
|------|-------|----------------------------------------------------------|-----------|
| 3:46 | Chang | One... two... breathe out... three... slow down... four. |           |
| 3:51 | Chang | Don't lean toward the jug. Stay square.                  |           |
| 4:04 | Chang | Breathe out as you stand.                                |           |

**Cues:** Hips back. Long back. Don't lean toward the jug. Stay square.

**Easier:** A lighter jug, and lift it from a higher surface like the counter's edge of a table. **Harder:** A heavier bag, or lift from the bottom step instead of the chair.

**Safety callouts:** Osteoporosis: this is the safe bending pattern. Keep the back long, never curl. Osteoporosis or a past spine fracture: long back always, never round to reach.

## 4:30–5:30 • Standing bird-dog

| Time | Who   | Spoken                                                                                         | On screen                         |
|------|-------|------------------------------------------------------------------------------------------------|-----------------------------------|
| 4:30 | Chang | Hand on the counter. Chair behind you.                                                         | Standing bird-dog                 |
| 4:42 | Chang | Slide your right foot back and lift it a few inches behind you, toes on or just off the floor. | Easier: Leg only, both hands down |
| 4:52 | Chang | Long back. No arching.                                                                         |                                   |
| 5:01 | Chang | Squeeze the buttock of the moving leg.                                                         |                                   |
| 5:11 | Chang | Breathe out as you reach.                                                                      |                                   |

**Cues:** Long back. No arching. Squeeze the buttock of the moving leg.

**Easier:** Leg only; both hands stay on the counter. **Harder:** Hold the reach for three seconds.

## 5:30–6:30 • Side lean at the counter

| Time | Who   | Spoken                                                              | On screen                      |
|------|-------|---------------------------------------------------------------------|--------------------------------|
| 5:30 | Chang | Hand on the counter. Chair behind you.                              | Side lean at the counter       |
| 5:42 | Chang | Lean your body into your forearm in a straight line, feet together. | Easier: Closer to counter, 5 s |
| 5:52 | Chang | Straight line from head to heels.                                   |                                |
| 6:01 | Chang | Keep talking. Your breathing never stops.                           |                                |
| 6:11 | Chang | Breathe normally the whole time. Count out loud.                    |                                |

**Cues:** Straight line from head to heels. Keep talking. Your breathing never stops. **Easier:** Stand closer to the counter, or hold for 5 seconds. **Harder:** Step farther away, or hold up to 20 seconds.

**Safety callouts:** Breathe through it. High blood pressure or a heart condition? Ask your doctor before long holds.

## 6:30–7:30 • Bridge on the bed

| Time | Who   | Spoken                                                   | On screen                       |
|------|-------|----------------------------------------------------------|---------------------------------|
| 6:30 | Chang | On your bed. Sit down first, then lie back slowly.       | Bridge on the bed               |
| 6:42 | Chang | Squeeze your buttocks and lift your hips a few inches.   | Easier: Squeeze without lifting |
| 6:52 | Chang | Push through your heels.                                 |                                 |
| 6:57 | Chang | One... two... breathe out... three... slow down... four. |                                 |
| 7:01 | Chang | Lift with your buttocks, not your back.                  |                                 |

| Time | Who   | Spoken                   | On screen |
|------|-------|--------------------------|-----------|
| 7:11 | Chang | Breathe out as you lift. |           |

**Cues:** Push through your heels. Lift with your buttocks, not your back.

**Easier:** Just squeeze your buttocks without lifting. **Harder:** Hold three seconds at the top, or lift higher.

**Safety callouts:** Sit a moment before you stand, then stand slowly while holding something steady.

## 7:30–8:00 • Stand up slowly

| Time | Who   | Spoken                                                               | On screen                     |
|------|-------|----------------------------------------------------------------------|-------------------------------|
| 7:30 | Chang | Roll to your side. Sit up. Ankle pumps.                              | Stand up slowly               |
| 7:41 | Chang | Sit a moment before you stand. Then stand slowly, holding something. | Easier: Smaller movements     |
| 7:49 | Chang | Breathe slowly.                                                      | Sit a moment before you stand |

**Cues:** Big, slow movements. Feel your calves work. **Easier:** Smaller movements. **Harder:** Lift one foot off the floor while you circle.

**Safety callouts:** Great before you stand up from sitting or lying. Sit a moment before you stand, then stand slowly while holding something steady.

## 8:00–8:50 • 4-6 breathing

| Time | Who   | Spoken                                         | On screen                   |
|------|-------|------------------------------------------------|-----------------------------|
| 8:00 | Chang | Sit tall. Chair against the wall.              | 4-6 breathing               |
| 8:11 | Chang | Breathe in through your nose for a count of 4. | Easier: In for 3, out for 4 |

| Time | Who   | Spoken                                                           | On screen |
|------|-------|------------------------------------------------------------------|-----------|
| 8:19 | Chang | Belly rises first.                                               |           |
| 8:27 | Chang | Soft shoulders.                                                  |           |
| 8:35 | Chang | In for 4, out for 6. That is about 6 breaths a minute. No holds. |           |

**Cues:** Belly rises first. Soft shoulders. **Easier:** In for 3, out for 4. **Harder:** Continue for 5 minutes.

**Safety callouts:** If you feel light-headed, go back to normal breathing.

### 8:50–9:15 • Close

| Time | Who   | Spoken                                                                      | On screen |
|------|-------|-----------------------------------------------------------------------------|-----------|
| 8:50 | Chang | You lifted more with a long back. That's the whole program in one sentence. |           |
| 9:03 | Chang | Walk today.                                                                 |           |

# BKS-B2 • Phase 2 (Build): Carry and walk

| Type                | Length | Set      | Wardrobe  | Voices |
|---------------------|--------|----------|-----------|--------|
| Back Strong program | 9:25   | SET-YARD | C-TRAIN-B | Chang  |

**On screen the whole session:** the "AI character" corner tag. In the first 30 seconds and before each hard block: "Stop if chest pain, dizziness or sharp pain."

**Gentle-day swaps (the sore knee / sore back / low energy button):** **Sore knee:** Use the Rebuild version of any standing leg move; skip floor practice and stairs today. **Sore back:** Seated versions of hinges and carries; keep every bend long-backed. **Low energy:** Do the warm-up, the first two blocks and the cool-down.

## Run of show

| Time      | Segment                                           | Rebuild             | Steady   | Strong                  | Iron                    |
|-----------|---------------------------------------------------|---------------------|----------|-------------------------|-------------------------|
| 0:00–0:35 | <b>Welcome</b>                                    | All levels together |          |                         |                         |
| 0:35–1:20 | <b>Pelvic tilt on the bed</b>                     | 6 small tilts       | 10 tilts | 10 tilts, 3-second hold | 12 tilts, 5-second hold |
| 1:20–1:50 | <b>Stand up slowly</b><br>Ankle pumps and circles | 10 pumps            | 10 pumps | 10 pumps                | 10 pumps                |

| Time      | Segment                                              | Rebuild                                              | Steady                   | Strong                               | Iron                                  |
|-----------|------------------------------------------------------|------------------------------------------------------|--------------------------|--------------------------------------|---------------------------------------|
| 1:50–3:00 | <b>Band press-out</b><br>Band press-out (anti-twist) | 2 sets of 8 each side, seated                        | 3 sets of 8 each side    | 3 sets of 8 each side, 5-second hold | 3 sets of 10 each side, 5-second hold |
| 3:00–4:15 | <b>One-side carry</b>                                | <i>Towel wring (grip):</i> 3 wrings each way         | 2 × 30 seconds each side | 3 × 30 seconds each side, heavier    | 3 × 45 seconds each side, heavy       |
| 4:15–8:15 | <b>Walk</b><br>Walk and talk                         | <i>Seated march:</i> 4 minutes: 1 on, 30 seconds off | 4 minutes easy           | 4 minutes, easy to brisk             | 4 minutes brisk                       |
| 8:15–9:00 | <b>Standing hip-front stretch</b>                    | 20 seconds each side, both hands down                | 20 seconds each side     | 20 seconds each side                 | 30 seconds each side                  |
| 9:00–9:25 | <b>Close</b>                                         | All levels together                                  |                          |                                      |                                       |

## Script

### 0:00–0:35 • Welcome

| Time | Who   | Spoken                                                                                                                                                                                                | On screen                                   |
|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 0:00 | Chang | Phase two. Carry, then walk.                                                                                                                                                                          | Back Strong · B2                            |
| 0:10 | Chang | In a big trial of adults who'd recently recovered from low back pain, people who followed a walking program plus education sessions went longer before it came back: a median of 208 days versus 112. | Stop if chest pain, dizziness or sharp pain |

| Time | Who   | Spoken                                                                | On screen                                        |
|------|-------|-----------------------------------------------------------------------|--------------------------------------------------|
| 0:17 | Chang | Walking is part of this program on purpose.                           | Study: 701 adults, walking + education, 6 months |
| 0:24 | Chang | Chest pain, dizziness or sharp pain: stop, sit, and call your doctor. |                                                  |

## 0:35–1:20 • Pelvic tilt on the bed

| Time | Who   | Spoken                                                                     | On screen                        |
|------|-------|----------------------------------------------------------------------------|----------------------------------|
| 0:35 | Chang | On your bed. Sit down first, then lie back slowly.                         | Pelvic tilt on the bed           |
| 0:45 | Chang | Gently flatten your lower back into the mattress by tightening your belly. | Easier: Smaller, one-second hold |
| 0:52 | Chang | Small movement.                                                            |                                  |
| 0:59 | Chang | Belly tightens, bottom stays down.                                         |                                  |
| 1:06 | Chang | Breathe out as you flatten.                                                |                                  |

**Cues:** Small movement. Belly tightens, bottom stays down. **Easier:** An even smaller movement, one-second hold. **Harder:** Hold five seconds.

**Safety callouts:** Osteoporosis or a past spine fracture: keep it small and gentle. Sit a moment before you stand, then stand slowly while holding something steady.

## 1:20–1:50 • Stand up slowly

| Time | Who   | Spoken               | On screen       |
|------|-------|----------------------|-----------------|
| 1:20 | Chang | Sit up. Ankle pumps. | Stand up slowly |

| Time | Who   | Spoken                                              | On screen                     |
|------|-------|-----------------------------------------------------|-------------------------------|
| 1:31 | Chang | Sit a moment, then stand slowly, holding something. | Easier: Smaller movements     |
| 1:39 | Chang | Breathe slowly.                                     | Sit a moment before you stand |

**Cues:** Big, slow movements. Feel your calves work. **Easier:** Smaller movements. **Harder:** Lift one foot off the floor while you circle.

**Safety callouts:** Great before you stand up from sitting or lying. Sit a moment before you stand, then stand slowly while holding something steady.

## 1:50–3:00 • Band press-out

| Time | Who   | Spoken                                                        | On screen                          |
|------|-------|---------------------------------------------------------------|------------------------------------|
| 1:50 | Chang | Sit or stand with the chair behind you. Check the band first. | Band press-out                     |
| 2:04 | Chang | Press your hands straight out in front of your chest.         | Easier: Seated, closer to the door |
| 2:15 | Chang | Stay square. The band pulls, you don't turn.                  |                                    |
| 2:21 | Chang | One... two... breathe out... three... slow down... four.      |                                    |
| 2:26 | Chang | Ribs down.                                                    |                                    |
| 2:37 | Chang | Breathe out as you press.                                     |                                    |

**Cues:** Stay square. The band pulls, you don't turn. Ribs down. **Easier:** Do it seated, or stand closer to the door. **Harder:** Step farther from the door, or hold 5 seconds.

**Safety callouts:** Door must open away from you and be locked.

## 3:00–4:15 • One-side carry

| Time | Who   | Spoken                                          | On screen                            |
|------|-------|-------------------------------------------------|--------------------------------------|
| 3:00 | Chang | Clear path. Free hand trails along the counter. | One-side carry                       |
| 3:15 | Chang | Stand tall. Don't lean away from the bag.       | Easier: Lighter bag, hand on counter |
| 3:27 | Chang | Tall like a lamppost.                           |                                      |
| 3:39 | Chang | Shoulders level.                                |                                      |
| 3:51 | Chang | Breathe steadily. You should be able to talk.   |                                      |

**Cues:** Tall like a lamppost. Shoulders level. **Easier:** A lighter bag, free hand on the counter the whole time. **Harder:** A heavier bag, or a longer walk.

**Safety callouts:** Hand arthritis flaring today? Squeeze gently, or skip and do the finger spread only. Pick up and put down with a hinge. Never round your back to lift.

## 4:15–8:15 • Walk

| Time | Who   | Spoken                                                           | On screen                             |
|------|-------|------------------------------------------------------------------|---------------------------------------|
| 4:15 | Chang | Good shoes. Cane or walker if you use one.                       | Walk                                  |
| 4:56 | Chang | Good shoes. Out the door or down the hallway.                    | Easier: Indoors, touching the counter |
| 5:34 | Chang | Tall. Arms swing. Easy pace to start.                            |                                       |
| 6:13 | Chang | Stiff at first? Often it eases as you warm up. Sharp pain, stop. |                                       |
| 6:51 | Chang | Breathe easy. You should be able to talk.                        |                                       |

**Cues:** Tall. Eyes ahead, not at your feet. Arms swing naturally. Heel, then toe. **Easier:** Walk indoors: kitchen to front door and back, touching the counter as you pass. Or march seated. **Harder:** Add 1-minute faster stretches, or a gentle hill.

**Safety callouts:** Chest pain, pressure, or unusual breathlessness: stop, sit, and call your doctor. If it doesn't settle in a few minutes, call 911.

## 8:15–9:00 • Standing hip-front stretch

| Time | Who   | Spoken                                                                     | On screen                             |
|------|-------|----------------------------------------------------------------------------|---------------------------------------|
| 8:15 | Chang | Hand on the counter. Chair behind you.                                     | Standing hip-front stretch            |
| 8:25 | Chang | Tuck your tailbone under slightly and squeeze the buttock of the back leg. | Easier: Smaller step, both hands down |
| 8:32 | Chang | Stay tall. Don't arch your back.                                           |                                       |
| 8:39 | Chang | Small movement, big squeeze.                                               |                                       |
| 8:46 | Chang | Breathe slowly.                                                            |                                       |

**Cues:** Stay tall. Don't arch your back. Small movement, big squeeze.

**Easier:** Smaller step. **Harder:** Reach the arm on the back-leg side up toward the ceiling, to comfortable height.

## 9:00–9:25 • Close

| Time | Who   | Spoken                                           | On screen |
|------|-------|--------------------------------------------------|-----------|
| 9:00 | Chang | Carry, walk, stretch. Your back likes all three. |           |
| 9:13 | Chang | Log your walking minutes.                        |           |

# BKS-A3 · Phase 3 (Strengthen): Confident bending

| Type                | Length | Set         | Wardrobe  | Voices |
|---------------------|--------|-------------|-----------|--------|
| Back Strong program | 9:25   | SET-KITCHEN | C-TRAIN-B | Chang  |

**On screen the whole session:** the "AI character" corner tag. In the first 30 seconds and before each hard block: "Stop if chest pain, dizziness or sharp pain."

**Gentle-day swaps (the sore knee / sore back / low energy button):** **Sore knee:** Use the Rebuild version of any standing leg move; skip floor practice and stairs today. **Sore back:** Seated versions of hinges and carries; keep every bend long-backed. **Low energy:** Do the warm-up, the first two blocks and the cool-down.

## Run of show

| Time      | Segment                                       | Rebuild             | Steady   | Strong   | Iron     |
|-----------|-----------------------------------------------|---------------------|----------|----------|----------|
| 0:00–0:35 | <b>Welcome</b>                                | All levels together |          |          |          |
| 0:35–1:00 | <b>Ankle pumps</b><br>Ankle pumps and circles | 10 pumps            | 10 pumps | 10 pumps | 10 pumps |

| Time          | Segment                                                      | Rebuild                                               | Steady                          | Strong                               | Iron                                   |
|---------------|--------------------------------------------------------------|-------------------------------------------------------|---------------------------------|--------------------------------------|----------------------------------------|
| 1:00–<br>1:40 | <b>March</b><br>Standing march at the counter                | <i>Seated march:</i><br>40 seconds                    | 40 seconds                      | 40 seconds                           | 40 seconds                             |
| 1:40–<br>2:00 | <b>Shoulder rolls</b>                                        | 6 rolls                                               | 8 rolls                         | 8 rolls                              | 8 rolls + circles                      |
| 2:00–<br>3:25 | <b>Suitcase lift</b><br>Suitcase lift (one-hand hinge)       | <i>Seated hip hinge:</i> 3 sets of 8, bottle at chest | 3 sets of 8 each side           | 3 sets of 8 each side, from the step | 3 sets of 10 each side, heavy bag      |
| 3:25–<br>4:40 | <b>Hinge pick-up</b><br>Counter hip hinge                    | <i>Seated hip hinge:</i> 2 sets of 10                 | 3 sets of 8, jug from the chair | 3 sets of 10, jug from the step      | 3 sets of 10, heavy bag from the step  |
| 4:40–<br>5:40 | <b>Side lean at the counter</b>                              | <i>Seated chest lift:</i> 10 lifts, 3-second hold     | 3 × 15 seconds each side        | 3 × 20 seconds each side             | 3 × 20 seconds each side, feet farther |
| 5:40–<br>6:40 | <b>Standing bird-dog</b><br>Standing bird-dog at the counter | 8 each side, leg only                                 | 10 each side, 3-second hold     | 12 each side, 3-second hold          | 12 each side, 5-second hold            |
| 6:40–<br>7:40 | <b>Two-hand carry</b><br>Grocery carry                       | <i>Towel wring (grip):</i> 4 wrings each way          | 2 × 40 seconds                  | 3 × 40 seconds, heavier              | 3 × 60 seconds, heavy                  |
| 7:40–<br>8:20 | <b>Calf stretch</b><br>Calf stretch at the wall              | 20 seconds each leg                                   | 20 seconds each leg             | 20 seconds each leg                  | 20 seconds each leg                    |
| 8:20–<br>9:00 | <b>Cyclic sighing</b>                                        | 30 seconds                                            | 40 seconds                      | 40 seconds                           | 40 seconds                             |

| Time      | Segment | Rebuild             | Steady | Strong | Iron |
|-----------|---------|---------------------|--------|--------|------|
| 9:00–9:25 | Close   | All levels together |        |        |      |

## Script

### 0:00–0:35 • Welcome

| Time | Who   | Spoken                                                                | On screen                                   |
|------|-------|-----------------------------------------------------------------------|---------------------------------------------|
| 0:00 | Chang | Phase three. Confident bending.                                       | Back Strong · A3                            |
| 0:10 | Chang | Pick up, put down, carry, turn. Like a real day.                      | Stop if chest pain, dizziness or sharp pain |
| 0:17 | Chang | Long back. Always.                                                    |                                             |
| 0:24 | Chang | Chest pain, dizziness or sharp pain: stop, sit, and call your doctor. |                                             |

### 0:35–1:00 • Ankle pumps

| Time | Who   | Spoken                                                            | On screen                 |
|------|-------|-------------------------------------------------------------------|---------------------------|
| 0:35 | Chang | Sit tall. Chair against the wall.                                 | Ankle pumps               |
| 0:42 | Chang | Point your toes away, then pull them toward you. That's one pump. | Easier: Smaller movements |
| 0:46 | Chang | Big, slow movements.                                              |                           |
| 0:50 | Chang | Feel your calves work.                                            |                           |
| 0:54 | Chang | Breathe slowly.                                                   |                           |

**Cues:** Big, slow movements. Feel your calves work. **Easier:** Smaller movements. **Harder:** Lift one foot off the floor while you circle.

**Safety callouts:** Great before you stand up from sitting or lying. Sit a moment before you stand, then stand slowly while holding something steady.

## 1:00–1:40 • March

| Time | Who   | Spoken                              | On screen             |
|------|-------|-------------------------------------|-----------------------|
| 1:00 | Chang | Stand slowly. Hand on the counter.  | March                 |
| 1:19 | Chang | March. Easy rhythm. Breathe steady. | Easier: Sit and march |

**Cues:** Stand tall. Knee up, foot down softly. **Easier:** Sit and do the seated march. **Harder:** Lift knees higher, march faster, or let one hand hover above the counter.

## 1:40–2:00 • Shoulder rolls

| Time | Who   | Spoken                                        | On screen                     |
|------|-------|-----------------------------------------------|-------------------------------|
| 1:40 | Chang | Sit tall. Chair against the wall.             | Shoulder rolls                |
| 1:46 | Chang | Lift your shoulders up toward your ears.      | Easier: Just shrug and let go |
| 1:49 | Chang | Big, slow circles, backward.                  |                               |
| 1:52 | Chang | Let your arms hang heavy.                     |                               |
| 1:55 | Chang | Breathe in as you lift, out as you roll down. |                               |

**Cues:** Big, slow circles, backward. Let your arms hang heavy. **Easier:** Just shrug up and let go. **Harder:** Add arm circles: small circles out to the sides, forward and back.

## 2:00–3:25 • Suitcase lift

| Time | Who   | Spoken                                                             | On screen                           |
|------|-------|--------------------------------------------------------------------|-------------------------------------|
| 2:00 | Chang | Hand on the counter. Chair behind you.                             | Suitcase lift                       |
| 2:16 | Chang | Push your hips back with a long back and bend your knees a little. | Easier: Lighter jug, higher surface |
| 2:30 | Chang | Hips back. Long back.                                              |                                     |
| 2:38 | Chang | One... two... breathe out... three... slow down... four.           |                                     |
| 2:43 | Chang | Don't lean toward the jug. Stay square.                            |                                     |
| 2:57 | Chang | Breathe out as you stand.                                          |                                     |

**Cues:** Hips back. Long back. Don't lean toward the jug. Stay square.

**Easier:** A lighter jug, and lift it from a higher surface like the counter's edge of a table. **Harder:** A heavier bag, or lift from the bottom step instead of the chair.

**Safety callouts:** Osteoporosis: this is the safe bending pattern. Keep the back long, never curl. Osteoporosis or a past spine fracture: long back always, never round to reach.

## 3:25–4:40 • Hinge pick-up

| Time | Who   | Spoken                                                         | On screen                              |
|------|-------|----------------------------------------------------------------|----------------------------------------|
| 3:25 | Chang | Hand on the counter. Chair behind you.                         | Hinge pick-up                          |
| 3:40 | Chang | Push your hips back, like closing a car door with your bottom. | Easier: Smaller hinge, hands stay down |
| 3:52 | Chang | Hips go back, not down.                                        |                                        |

| Time | Who   | Spoken                                                   | On screen |
|------|-------|----------------------------------------------------------|-----------|
| 3:58 | Chang | One... two... breathe out... three... slow down... four. |           |
| 4:04 | Chang | Long back, chest proud. No rounding.                     |           |
| 4:16 | Chang | Breathe out as you stand tall.                           |           |

**Cues:** Hips go back, not down. Long back, chest proud. No rounding. Stand up by squeezing your buttocks. **Easier:** Smaller movement, hands stay on the counter the whole time. **Harder:** Pick up a jug of water from the seat of a chair using the same hinge, and set it back down. Later: from a lower surface, like a step.

**Safety callouts:** Osteoporosis: this is the safe bending pattern. Keep the back long, never curl. Osteoporosis or a past spine fracture: this is the safe way to bend. Keep your back long; never round forward under load.

## 4:40–5:40 • Side lean at the counter

| Time | Who   | Spoken                                           | On screen                      |
|------|-------|--------------------------------------------------|--------------------------------|
| 4:40 | Chang | Forearm on the counter. Lean in a straight line. | Side lean at the counter       |
| 4:59 | Chang | Keep talking. Count out loud.                    | Easier: Closer to counter, 5 s |
| 5:15 | Chang | High blood pressure: keep holds short.           |                                |

**Cues:** Straight line from head to heels. Keep talking. Your breathing never stops. **Easier:** Stand closer to the counter, or hold for 5 seconds. **Harder:** Step farther away, or hold up to 20 seconds.

**Safety callouts:** Breathe through it. High blood pressure or a heart condition? Ask your doctor before long holds.

## 5:40–6:40 • Standing bird-dog

| Time | Who   | Spoken                                                                                         | On screen                         |
|------|-------|------------------------------------------------------------------------------------------------|-----------------------------------|
| 5:40 | Chang | Hand on the counter. Chair behind you.                                                         | Standing bird-dog                 |
| 5:52 | Chang | Slide your right foot back and lift it a few inches behind you, toes on or just off the floor. | Easier: Leg only, both hands down |
| 6:02 | Chang | Long back. No arching.                                                                         |                                   |
| 6:11 | Chang | Squeeze the buttock of the moving leg.                                                         |                                   |
| 6:21 | Chang | Breathe out as you reach.                                                                      |                                   |

**Cues:** Long back. No arching. Squeeze the buttock of the moving leg.

**Easier:** Leg only; both hands stay on the counter. **Harder:** Hold the reach for three seconds.

## 6:40–7:40 • Two-hand carry

| Time | Who   | Spoken                                                | On screen                              |
|------|-------|-------------------------------------------------------|----------------------------------------|
| 6:40 | Chang | Clear path. Walk along the counter or a hallway wall. | Two-hand carry                         |
| 6:52 | Chang | Stand tall, shoulders down.                           | Easier: One light bag, hand on counter |
| 7:02 | Chang | Tall, like you're proud of the groceries.             |                                        |
| 7:11 | Chang | Short, steady steps.                                  |                                        |
| 7:21 | Chang | Breathe steadily. You should be able to talk.         |                                        |

**Cues:** Tall, like you're proud of the groceries. Short, steady steps. **Easier:** One light bag, and keep your free hand trailing along the counter. **Harder:** Heavier: two full gallon jugs (about 8 pounds each). Iron track: two heavier bags or kettlebells.

**Safety callouts:** Hand arthritis flaring today? Squeeze gently, or skip and do the finger spread only. Pick up and put down with a hinge. Never round your back to lift.

## 7:40–8:20 • Calf stretch

| Time | Who   | Spoken                                                                                         | On screen                 |
|------|-------|------------------------------------------------------------------------------------------------|---------------------------|
| 7:40 | Chang | Hand on the counter. Chair behind you.                                                         | Calf stretch              |
| 7:49 | Chang | Bend the front knee and lean gently toward the wall until you feel a stretch in the back calf. | Easier: Smaller step back |
| 7:55 | Chang | Back heel stays down.                                                                          |                           |
| 8:02 | Chang | Toes point forward.                                                                            |                           |
| 8:08 | Chang | Breathe slowly through the stretch.                                                            |                           |

**Cues:** Back heel stays down. Toes point forward. **Easier:** Smaller step back. **Harder:** Bigger step back, or bend the back knee slightly to reach the lower calf.

## 8:20–9:00 • Cyclic sighing

| Time | Who   | Spoken                                                  | On screen                        |
|------|-------|---------------------------------------------------------|----------------------------------|
| 8:20 | Chang | Sit. Two sniffs in through the nose. One long sigh out. | Cyclic sighing                   |
| 8:39 | Chang | No holding.                                             | Easier: Just slow nose breathing |

**Cues:** Two sniffs in, one long sigh out. No holding at any point. **Easier:** Just breathe slowly through your nose, making the out-breath a little longer. **Harder:** Build up to 5 minutes.

**Safety callouts:** If you feel light-headed, return to normal breathing.

## 9:00–9:25 • Close

| Time | Who   | Spoken                                                 | On screen |
|------|-------|--------------------------------------------------------|-----------|
| 9:00 | Chang | Twelve weeks of long-back bending. That's a habit now. |           |
| 9:13 | Chang | Final checkpoint this weekend.                         |           |

# BKS-B3 · Phase 3 (Strengthen): Turn, carry, walk

| Type                | Length | Set      | Wardrobe  | Voices |
|---------------------|--------|----------|-----------|--------|
| Back Strong program | 10:15  | SET-PROM | C-TRAIN-B | Chang  |

**On screen the whole session:** the "AI character" corner tag. In the first 30 seconds and before each hard block: "Stop if chest pain, dizziness or sharp pain."

**Gentle-day swaps (the sore knee / sore back / low energy button):** **Sore knee:** Use the Rebuild version of any standing leg move; skip floor practice and stairs today. **Sore back:** Seated versions of hinges and carries; keep every bend long-backed. **Low energy:** Do the warm-up, the first two blocks and the cool-down.

## Run of show

| Time      | Segment                                | Rebuild             | Steady      | Strong                 | Iron                    |
|-----------|----------------------------------------|---------------------|-------------|------------------------|-------------------------|
| 0:00–0:35 | <b>Welcome</b>                         | All levels together |             |                        |                         |
| 0:35–1:20 | <b>Gentle seated turn</b>              | 4 each side, small  | 6 each side | 8 each side            | 8 each side             |
| 1:20–1:55 | <b>Chest lift</b><br>Seated chest lift | 6 lifts             | 8 lifts     | 8 lifts, 3-second hold | 10 lifts, 3-second hold |

| Time       | Segment                                                     | Rebuild                                              | Steady                               | Strong                                | Iron                                  |
|------------|-------------------------------------------------------------|------------------------------------------------------|--------------------------------------|---------------------------------------|---------------------------------------|
| 1:55–3:00  | <b>Band press-out</b><br>Band press-out (anti-twist)        | 2 sets of 8 each side, seated                        | 3 sets of 8 each side, 5-second hold | 3 sets of 10 each side, 5-second hold | 3 sets of 12 each side, 5-second hold |
| 3:00–4:10  | <b>One-side carry</b>                                       | <i>Towel wring (grip):</i> 4 wrings each way         | 3 × 30 seconds each side             | 3 × 45 seconds each side              | 3 × 60 seconds each side, heavy       |
| 4:10–9:10  | <b>Walk</b><br>Walk and talk                                | <i>Seated march:</i> 5 minutes: 1 on, 30 seconds off | 5 minutes steady                     | 5 minutes brisk                       | 5 minutes brisk, hills                |
| 9:10–9:50  | <b>Calf stretch at the rail</b><br>Calf stretch at the wall | 20 seconds each leg                                  | 20 seconds each leg                  | 20 seconds each leg                   | 20 seconds each leg                   |
| 9:50–10:15 | <b>Close</b>                                                | All levels together                                  |                                      |                                       |                                       |

## Script

### 0:00–0:35 • Welcome

| Time | Who   | Spoken                                                                | On screen                                   |
|------|-------|-----------------------------------------------------------------------|---------------------------------------------|
| 0:00 | Chang | Phase three. Turn, carry, walk.                                       | Back Strong · B3                            |
| 0:10 | Chang | Longer walk today.                                                    | Stop if chest pain, dizziness or sharp pain |
| 0:17 | Chang | Good shoes. Flat route.                                               |                                             |
| 0:24 | Chang | Chest pain, dizziness or sharp pain: stop, sit, and call your doctor. |                                             |

## 0:35–1:20 • Gentle seated turn

| Time | Who   | Spoken                                                                 | On screen             |
|------|-------|------------------------------------------------------------------------|-----------------------|
| 0:35 | Chang | Sit tall. Chair against the wall.                                      | Gentle seated turn    |
| 0:45 | Chang | Turn your chest to the right, only as far as comfortable. Eyes follow. | Easier: Smaller turns |
| 0:52 | Chang | Grow tall first, then turn.                                            |                       |
| 0:59 | Chang | Hips stay still.                                                       |                       |
| 1:06 | Chang | Breathe out as you turn.                                               |                       |

**Cues:** Grow tall first, then turn. Hips stay still. **Easier:** Smaller turns.

**Harder:** Place one hand on the opposite knee as a light guide. Never pull into the turn.

**Safety callouts:** Osteoporosis? Turn gently and only as far as comfortable. Never force the twist.

## 1:20–1:55 • Chest lift

| Time | Who   | Spoken                                               | On screen            |
|------|-------|------------------------------------------------------|----------------------|
| 1:20 | Chang | Sit tall. Chair against the wall.                    | Chest lift           |
| 1:28 | Chang | Lift your chest up toward the ceiling, growing tall. | Easier: Smaller lift |
| 1:34 | Chang | Grow tall.                                           |                      |
| 1:39 | Chang | Look straight ahead, not up.                         |                      |
| 1:45 | Chang | Breathe in as you lift, out as you relax.            |                      |

**Cues:** Grow tall. Look straight ahead, not up. **Easier:** Smaller lift. **Harder:** Hold three seconds at the top.

## 1:55–3:00 • Band press-out

| Time | Who   | Spoken                                                        | On screen                          |
|------|-------|---------------------------------------------------------------|------------------------------------|
| 1:55 | Chang | Sit or stand with the chair behind you. Check the band first. | Band press-out                     |
| 2:08 | Chang | Press your hands straight out in front of your chest.         | Easier: Seated, closer to the door |
| 2:18 | Chang | Stay square. The band pulls, you don't turn.                  |                                    |
| 2:24 | Chang | One... two... breathe out... three... slow down... four.      |                                    |
| 2:29 | Chang | Ribs down.                                                    |                                    |
| 2:39 | Chang | Breathe out as you press.                                     |                                    |

**Cues:** Stay square. The band pulls, you don't turn. Ribs down. **Easier:** Do it seated, or stand closer to the door. **Harder:** Step farther from the door, or hold 5 seconds.

**Safety callouts:** Door must open away from you and be locked.

## 3:00–4:10 • One-side carry

| Time | Who   | Spoken                                          | On screen                            |
|------|-------|-------------------------------------------------|--------------------------------------|
| 3:00 | Chang | Clear path. Free hand trails along the counter. | One-side carry                       |
| 3:14 | Chang | Stand tall. Don't lean away from the bag.       | Easier: Lighter bag, hand on counter |
| 3:25 | Chang | Tall like a lamppost.                           |                                      |
| 3:36 | Chang | Shoulders level.                                |                                      |
| 3:47 | Chang | Breathe steadily. You should be able to talk.   |                                      |

**Cues:** Tall like a lamppost. Shoulders level. **Easier:** A lighter bag, free hand on the counter the whole time. **Harder:** A heavier bag, or a longer walk.

**Safety callouts:** Hand arthritis flaring today? Squeeze gently, or skip and do the finger spread only. Pick up and put down with a hinge. Never round your back to lift.

## 4:10–9:10 • Walk

| Time | Who   | Spoken                                                                          | On screen                             |
|------|-------|---------------------------------------------------------------------------------|---------------------------------------|
| 4:10 | Chang | Good shoes. Cane or walker if you use one.                                      | Walk                                  |
| 5:13 | Chang | Out the door or down the hallway. Tall. Arms swing.                             | Easier: Indoors, touching the counter |
| 6:13 | Chang | Steady pace. Can talk, can't sing.                                              |                                       |
| 7:13 | Chang | Breathe easy. Notice how your back feels at minute five compared with week one. |                                       |

**Cues:** Tall. Eyes ahead, not at your feet. Arms swing naturally. Heel, then toe. **Easier:** Walk indoors: kitchen to front door and back, touching the counter as you pass. Or march seated. **Harder:** Add 1-minute faster stretches, or a gentle hill.

**Safety callouts:** Chest pain, pressure, or unusual breathlessness: stop, sit, and call your doctor. If it doesn't settle in a few minutes, call 911.

## 9:10–9:50 • Calf stretch at the rail

| Time | Who   | Spoken                                 | On screen                |
|------|-------|----------------------------------------|--------------------------|
| 9:10 | Chang | Hand on the counter. Chair behind you. | Calf stretch at the rail |

| Time | Who   | Spoken                                                                                         | On screen                 |
|------|-------|------------------------------------------------------------------------------------------------|---------------------------|
| 9:19 | Chang | Bend the front knee and lean gently toward the wall until you feel a stretch in the back calf. | Easier: Smaller step back |
| 9:25 | Chang | Back heel stays down.                                                                          |                           |
| 9:32 | Chang | Toes point forward.                                                                            |                           |
| 9:38 | Chang | Breathe slowly through the stretch.                                                            |                           |

**Cues:** Back heel stays down. Toes point forward. **Easier:** Smaller step back. **Harder:** Bigger step back, or bend the back knee slightly to reach the lower calf.

## 9:50–10:15 • Close

| Time  | Who   | Spoken                                                | On screen |
|-------|-------|-------------------------------------------------------|-----------|
| 9:50  | Chang | Twelve weeks. Stronger back, more walking. Keep both. |           |
| 10:03 | Chang | Final checkpoint this weekend.                        |           |

# Where the numbers come from

Every number and health statement in this book traces to one of the entries below. They are the same evidence IDs our whole team uses (EVIDENCE.md). Grade A means a Cochrane review, a large meta-analysis or a guideline body. B means a single good trial or a small meta-analysis. C means an observational study: it shows a link, not cause and effect. D means lab, physiology or label data.

**E22** (grade C). Daily steps: in adults 60+, mortality benefit plateaued at about 6,000–8,000 steps/day. The top quartiles had 40–53% lower risk than the lowest (about 3,500/day).

Source: Paluch et al., Lancet Public Health 2022 — <https://www.sciencedaily.com/releases/2022/03/220303112207.htm>

**E30** (grade A). Exercise for chronic low back pain: 249 trials, 24,486 people. Pain about 15 points better (0–100 scale) at 3 months vs no treatment or usual care. Function improved modestly.

Source: Hayden et al., Cochrane 2021 — [https://www.cochrane.org/evidence/CD009790\\_exercise-treatment-chronic-low-back-pain](https://www.cochrane.org/evidence/CD009790_exercise-treatment-chronic-low-back-pain)

**E30b** (grade A). Yoga for chronic non-specific low back pain gives small-to-moderate improvements in function vs no exercise.

Source: Wieland et al., Cochrane 2022 — <https://www.cochranelibrary.com/cdsr/doi/10.1002/14651858.CD010671.pub3>

**E40** (grade A (consensus)). Osteoporosis exercise guidance: do resistance and balance training. Avoid rapid, repeated or loaded spinal flexion (sit-ups, toe-touches with weight) and forceful twisting. Flexion-exercise programs were linked with more vertebral fractures.

Source: Giangregorio et al., "Too Fit To Fracture", Osteoporos Int 2014 — <https://link.springer.com/10.1007/s00198-014-2881-4> ; Sinaki & Mikkelsen, Arch Phys Med Rehabil 1984

**E41** (grade B (known literature)). Pain-monitoring model: during and after exercise, pain up to about 3/10 (we use a conservative threshold) is acceptable if it settles by the next morning and isn't trending up week to week.

Source: Thomeé, Phys Ther 1997; Silbernagel et al., Am J Sports Med 2007

**E42** (grade A (statement)). AHA scientific statement (2023): resistance training is recommended for adults with and without cardiovascular disease. It lowers blood pressure and improves glucose control and function. Avoid breath-holding (Valsalva). People with CVD should get individualized clearance.

Source: Paluch et al., Circulation 2024 — <https://www.ahajournals.org/doi/10.1161/CIR.0000000000001189>

**E43** (grade A (guideline)). Pre-exercise screening (ACSM 2015): anyone with symptoms of cardiovascular, metabolic or renal disease (chest pain, unusual breathlessness, fainting) should get medical clearance before starting. Otherwise, light-to-moderate exercise can begin and progress.

Source: Riebe et al., Med Sci Sports Exerc 2015 (known literature)

**E46** (grade A– (meta, known literature)). Orthostatic hypotension (a blood-pressure drop on standing) is common in older adults. A meta-analysis found it in about 22% of community-dwelling older people. The risk is higher with some blood-pressure and other medicines. Practical rule: sit, pump your ankles, then stand slowly while holding something.

Source: Saedon et al., J Gerontol A Biol Sci Med Sci 2020 (prevalence of orthostatic hypotension, systematic review & meta-analysis)

**E60** (grade B+ (single large RCT)). WalkBack RCT: 701 adults who had recently recovered from an episode of low back pain. An individualised, progressive walking program plus six physiotherapist-guided education sessions over 6 months: median 208 days before a recurrence vs 112 days in the no-intervention group; healthcare visits and work absence roughly halved. Say "people who walked... went longer before their back pain came back", never "walking prevents back pain". Not an older-adult-only sample.

Source: Pocovi et al., Lancet 2024 — opened summary <https://www.sciencedaily.com/releases/2024/06/240620152321.htm> (primary: [https://www.thelancet.com/journals/lancet/article/PIIS0140-6736\(24\)00755-4/fulltext](https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(24)00755-4/fulltext), not opened)